Association testing II

```
$ echo "Data Sciences Institute"
```

What You'll Learn Today

- **Modeling association with population structure:**
  - When and how to use linear mixed models to handle relatedness and population structure.
- **Discoveries at scale:**
  - How to control error rates across millions of tests (Bonferroni genome-wide threshold, FDR).
- **Combine evidence across studies:**
  - GWAS meta-analysis (combine effect sizes; fixed vs. random effects)

Mixed Effect models

- Handle population structure, family relatedness, and cryptic correlations
- Particularly useful when measurements are made on clusters or related individuals (family).
- Model phenotypes using a mixture of fixed effects (SNPs, covariates) and random effects (family structure).

Linear Mixed Models (LMM)

- Basic linear model: $Y = X\beta + C\alpha + \varepsilon$
- Mixed model extension: $Y = X\beta + C\alpha + u + \varepsilon$
  - u : genetic random effects
  - ε : residual, non-heritable variation
- Assumptions on random effects: $E(u) = 0, \quad \mathrm{Var}(u) = \sigma_g^2 K.$
- In practice, K is approximated as $K = GG^T/M$, where G is the genotype matrix constructed from all M SNPs.
- Alternatively,K can be estimated from family pedigree data.

Linear Mixed Models (LMM)

- K captures genetic relatedness, including population structure, family relationships, and hidden relatedness
- σ_g^2: genetic variance parameter we aim to estimate
- Estimation methods: REML (Restricted Maximum Likelihood) or AI-REML (Average Information REML)
- LMMs also allow us to estimate the individual random effects(μ)

Association testing in the LMM framework

Two-step fitting procedure for LMM:

- STEP 1: fit the null model.

$$Y = C\alpha + u + \varepsilon$$

- $E(u) = 0, \mathrm{Var}(u) = \sigma_g^2 K, \mathrm{Var}(\varepsilon) = \sigma_e^2 I$
- Using REML/AI-REML we can estimate $\widehat{\sigma_g^2}$ and $\widehat{\sigma_e^2}$.
- We can also get BLUP (best linear unbiased predictors)

$$\widehat{u} = \mathrm{K}\widehat{\sigma_g^2} \times \Sigma^{-1}\left(\mathrm{I} - \mathrm{C}\left(C^T\Sigma^{-1}\mathrm{C}\right)^{-1} C^T\Sigma^{-1}\right)\mathrm{Y}, \Sigma = \widehat{\sigma_g^2}\,\mathrm{K} + \widehat{\sigma_e^2}\mathrm{I}.$$

Association testing in the LMM framework

- STEP 2: test for association with each SNP. $H_0 : \beta = 0$.

$$\widehat{\boldsymbol{Y}}^{\star}_{\text{resid}} = \boldsymbol{Y} - \widehat{u} - C\hat{\alpha}.$$

- Test for association in a linear (non-mixed) regression model

$$\widehat{\boldsymbol{Y}}^{\star}_{\text{resid}} = X\beta + \varepsilon$$

- STEP 1 is computationally demanding (large matrix operations like inversions), but it only needs to be done once under the null
- STEP 2 is then repeated for millions of SNPs

Heritability Estimation from GWAS

- Variance decomposition: $Var(Y) = Var(G) + Var(\varepsilon)$
- **Heritability:**$h^2 = \frac{Var(G)}{Var(Y)}$
- Earlier: used **trait covariance among relatives** (no genotypes)
- Now: estimate heritability directly from **GWAS data**
- $h^2 = \frac{\widehat{\sigma_g^2}}{\sigma_Y^2}$

Efficient LMM Methods for GWAS

- BOLT-LMM (Nature genetics, 2015)
  - Software: https://alkesgroup.broadinstitute.org/BOLT-LMM/BOLTLMMmanual.html
- fastGWA (Nature genetics, 2019) - assumes a sparse GRM
  - Software: https://yanglab.westlake.edu.cn/software/gcta/\#Overview
- SAIGE (Nature genetics, 2018)
  - Software: https://github.com/weizhouUMICH/SAIGE
- Regenie (Nature genetics, 2021)
  - Software: https://rgcgithub.github.io/regenie/

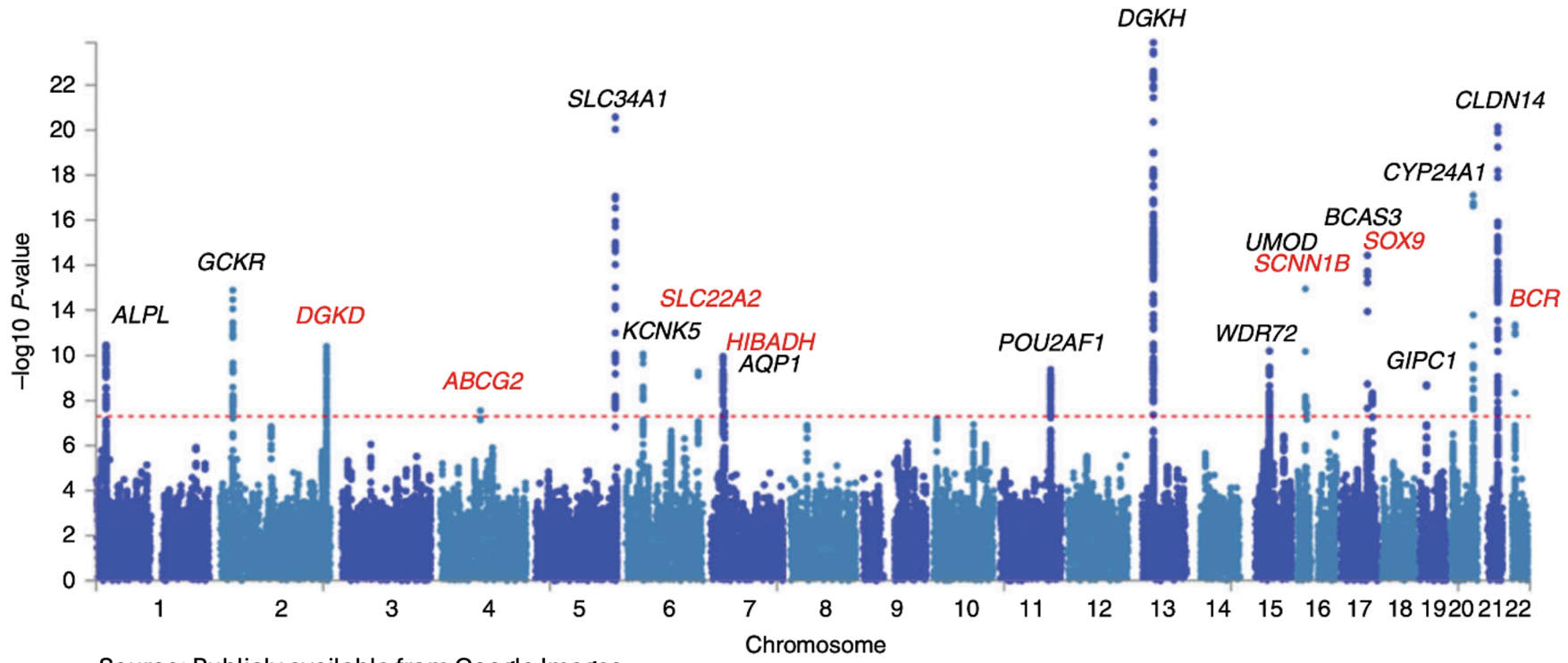
Complications when testing association with millions of markers in large GWAS studies

- So far, we have discussed one test/one genetic marker at a time.
- **Multiple testing in GWAS** - millions of tests at once
- **Meta-analysis of multiple datasets** - combine data/results from multiple studies

Multiple Testing

GWAS of Kidney Stone Disease

- In a typical GWAS we perform millions of tests. How do we account for that? What Significance level should we use?



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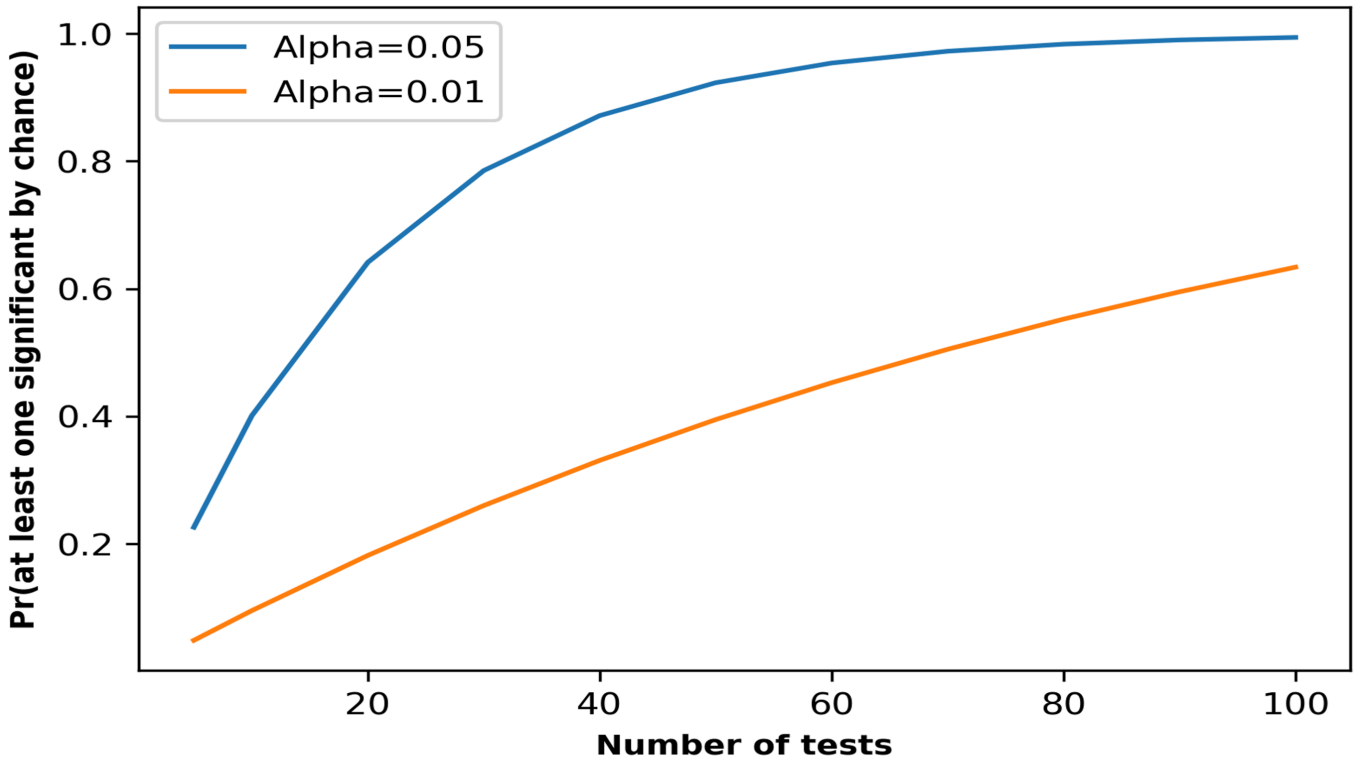
Multiple Testing

- Multiple testing issues arise when many markers are tested as in GWAS.
- Major statistical problem as it leads to increased false positive rates if not accounted for.
- Idea: Test each marker separately and adjust the significance level of each test.

Why do we need to adjust for multiple testing?



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Methods based on P-value Adjustment

- Test each SNP separately and adjust the significance level of each test in order to preserve the overall error rate.
- Two different error rates:
  i. Family-wise error rate (FWER)
  ii. False discovery rate (FDR)
- If M is the total number of tested SNPs, then for each $m = 1, \ldots, M$, we define the null hypothesis:
  $H_0(m)$: no association between the m-th SNP and the phenotype.

Bonferroni method

- **FWER (family-wise error rate)**:

$$FWER = P\left(\text{ reject at least one } H_0(m) \mid H_0(m) \text{ is true for all } m\right)$$

- **Bonferroni correction**: fix $FWER = \alpha$ and set individual significance levels at $\frac{\alpha}{M}$.
- This ensures that the FWER is less than the desired level α (e.g. 0.05).
- Bonferroni correction is conservative if markers are not independent (due to linkage disequilibrium).

Bonferroni correction

```r
# Desired family-wise error rate (FWER)
alpha <- 0.05

# Input p-values
p <- c(0.002, 0.01, 0.02, 0.03, 0.20)

# Total number of tests
M <- length(p)

# Bonferroni significance cutoff
alpha / M

# Test which p-values remain significant
p < alpha / M

# [1]  TRUE FALSE FALSE FALSE FALSE
```

Bonferroni threshold for GWAS

- The effective number of independent tests in a dense GWAS study is 1 million, so the Bonferroni adjustment corresponds to a significance level of 5×10^{-8}.
- This corresponds to a finding by chance 1 in 20 GWAS studies.
- Large sample sizes are needed for such a stringent threshold.

False Discovery Rate (FDR)

- Rather than control the Type-1 error, FDR limits the expected number of null-hypotheses that are rejected incorrectly.
- FDR=5% means that on average 5% of the SNPs we rejected are in fact false positives.
- FDR is less conservative than FWER $\rightarrow$ higher power.

False Discovery Rate (FDR)

	Declared non-significant	**Declared significant**	
True null hypotheses	U	V	M_0
False null hypotheses	T	S	$M - M_0$
	$M - R$	R	M

- Assume there are M independent markers.
- The false discovery rate is $\mathrm{E}\left(\frac{V}{R}\right)$ (expectation of false discovery proportion).
- Goal: keep the FDR below a specific threshold, e.g. 0.05 or 0.10.

Benjamini–Hochberg (BH) procedure

- Benjamini and Hochberg (1995) procedure can be used to control the FDR.
- Rank the M p-values from smallest to largest:

$$p_{(1)}, \ldots, p_{(M)}$$

- For a specified FDR level (e.g. 0.05), compare

$$\frac{M}{i} p_{(i)} \leq FDR$$

- Find largest i for which this inequality holds, and then reject tests that correspond to $1, \ldots, i$.
- For dependent tests, extensions are available (e.g. Benjamini-Yekutieli 2001).

Benjamini-Hochberg (BH) procedure

```r
# target false discovery rate (FDR)
alpha <- 0.05

# Input p-values
p <- c(0.002, 0.01, 0.02, 0.03, 0.20)

# BH-adjusted p-values
p_bh <- p.adjust(p, method = "BH")


# Which tests are significant after BH correction?
p_bh < alpha

# [1]  TRUE  TRUE  TRUE  TRUE FALSE
```

Exercise

We have p-values from 10 hypothesis tests:

```r
p <- c(0.002, 0.005, 0.006, 0.008, 0.009, 0.009, 0.017, 0.025, 0.105, 0.54)
```

At significance level $\alpha = 0.05$:

- Which tests are rejected using the Bonferroni correction?
- Which tests are rejected using the Benjamini–Hochberg (BH) procedure?

Meta-analysis

- Meta-analysis is essential in GWAS.
- GWAS studies are extremely large $\rightarrow$ require combining many smaller cohorts.
    - Example: the largest GWAS on height (2022) analyzed 5.4 million individuals across diverse ancestries and identified 12,111 independent SNPs.
- Purpose: to combine information from multiple independent studies.

Meta-analysis

- Combines information from multiple independent studies.
- Increases statistical power by boosting sample size.
- Helps assess consistency of findings across datasets.
- Methods:
  - Fixed-effects meta-analysis
  - Random-effects meta-analysis

Fixed-effects meta-analysis

- Constant effect size across studies.
- Observed effect size in each study varies due to random sampling error.
- **Combined effect estimates the fixed effect size (the same underlying parameter across studies)**.
- Might be realistic if, for example, the studies have all been conducted in the same population, consistently measured phenotypes, same inclusion criteria etc.
- In practice, that may not be true.

Fixed-effect model

- We assume we have K studies.
- Let $\hat{\beta}_1, \ldots, \hat{\beta}_K$ be the effect-size estimates (e.g. $\log(OR)$ or regression coefficients).
- Let $\hat{\sigma}_i = SE\left(\hat{\beta}_i\right)^2$ (within study variance) and $w_i = \hat{\sigma}_i{}^{-1}$.
- The overall inverse-variance-weighted effect-size is:

$$\hat{\beta} = \frac{\sum_{i=1}^{K} w_i \widehat{\beta}_i}{\sum_{i=1}^{K} w_i}.$$

Fixed-effect model

- $\hat{\beta}$ follows a normal distribution, with $\mathrm{SE}(\hat{\beta}) = 1/\sqrt{\sum_{i=1}^{K} w_i}$.

$$\hat{\beta} \sim \mathrm{N}\left(\beta, \left(\sum_{i=1}^{K} w_i\right)^{-1}\right)$$

- Larger studies are given higher weight compared with smaller studies. So if one very large study and other small studies, the large study will dominate.

Fixed-effect meta-analysis

```r
# Load package
library(metafor)

# Example study estimates and standard errors
beta  = c(0.05, 0.12, 0.18, 0.40, 0.52, 0.60)
se    = c(0.04, 0.05, 0.04, 0.05, 0.04, 0.05)
```

```r
# Fixed-effect meta-analysis
res_fe <- rma(yi = beta, sei = se, method = "FE")
summary(res_fe)
```

Random-effects meta analysis

Random-effects meta-analysis.

- True effect size may vary from study to study (so there is a study-specific true effect).
- Observed effect size in each study varies due to both random error and differences in true effect sizes across studies.
- **Combined effect estimates the mean of the distribution of true effects**.

Random-effect model

- In the RE model, we assume that the true effects $\beta_i = \beta + \xi_i, \xi_i \sim N\left(0, \tau^2\right)$.
- The overall effect-size is estimated as:

$$\hat{\beta} = \frac{\sum_{i=1}^{K} w_i^* \hat{\beta}_i}{\sum_{i=1}^{K} w_i^*}$$

- $w_i^* = \left(\hat{\sigma}_i + \hat{\tau}^2\right)^{-1}$.
- The weight here depends both on the within study variance and also on the between-study heterogeneity in true effects.
- $$\hat{\beta} \sim \mathrm{N}\left(\beta, \left(\sum_{i=1}^{K} w_i^*\right)^{-1}\right).$$

Random-effect model

- $w_i^* = \left(\hat{\sigma}_i + \hat{\tau}^2\right)^{-1}$ (in fixed effect $w_i = \hat{\sigma}_i{}^{-1}$)
- Smaller studies are given relatively more weight than in the fixed effect model
- The calculated standard error is smaller from a fixed effects meta-analysis than that from a random-effects meta-analysis.
- How to estimate the between-study variance $\hat{\tau}^2$?
- Many different methods, e.g. DerSimonian and Laird (DL)

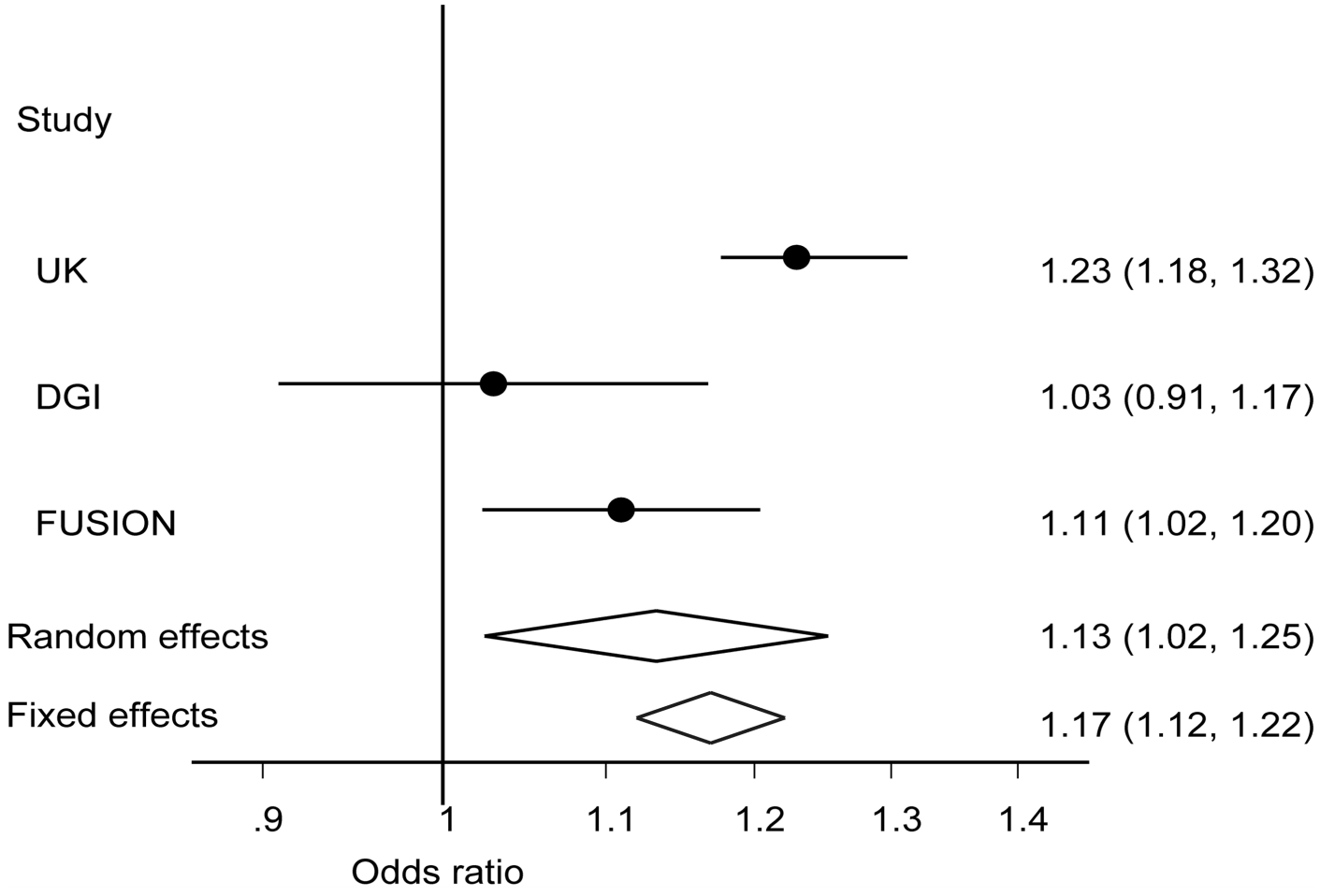
```r
# Load package
library(metafor)

# Example study estimates and standard errors
beta  = c(0.05, 0.12, 0.18, 0.40, 0.52, 0.60)
se    = c(0.04, 0.05, 0.04, 0.05, 0.04, 0.05)
```

```r
# Random-effects meta-analysis
rma(yi = beta, sei = se, method = "REML")
```

Fixed vs. Random Effect model

- The decision needs to be done before the analysis based on knowledge about the individual studies
- **Fixed effect meta-analysis is typically used in genetics**, without regard to heterogeneity.
- Random effect meta-analysis is typically too conservative (less powerful).
- Other GWAS meta-analysis tools
    - METAL: fixed-effect only
    - GWAMA: fixed-effect and random-effects

Meta-analysis of three GWAS studies assessing the link between the FTO rs8050136 variant and type 2 diabetes



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What's Next

- Population Stratification
- Genotype Imputation
- Quality Control

What questions do you have about anything from today?